

Dmitry Messerman

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Location: Haifa, Israel

Professional Summary

Ph. D. in Electrical Engineering

EDA CAD tool expert with over 20 years of experience in the development, support, and deployment of Computer Aided Design and Silicon Debug tools for high-speed and analog custom VLSI design.

Renowned for multi-disciplinary expertise, problem solving and algorithm development.

Work Experience

Principal Engineer at Intel Corporation, Haifa, Israel 1992 - 2025

- Led the development, deployment customization and support of in-house and commercial Electronic Design Automation tools in multiple areas:

- Modeling parasitic effects in VLSI design – has been used in production for ~10 years, patent granted
- Reliability validation of VLSI design – has been used in production for ~20 years
- Surrogate modeling of silicon heat dissipation in tester environment based on diffusion equations solver

- Led development of silicon debug tools

- Platform HW/SW co-simulation for BIOS Memory Reference Code validation
- Structure-Based Functional Tests (SBFT) coverage improvement based on genetic algorithm

- Development, customization and support of in-house block and full chip level VLSI layout editors

- Recognized for exceptional debugging and problem-solving capabilities in CAD software and Linux

Physicist and software engineer in FrantzTech Ltd 1990 – 1992

- Co-developed ultrasound image processing algorithm for early detection of diffused liver diseases

Researcher in Research Institute for High-Voltage Direct Current Power Transmission 1982-1990

- Analysis of the environmental impact of electric fields from extra-high-voltage power lines
- Development of fundamental algorithms for calculating the modal parameters of multi-conductor power transmission lines, taking into account the characteristics of real soil.
- Investigation of the distortion and attenuation of lightning surge waves in long-span power transmission lines

Education

Ph. D. in Electrical Engineering, St. Petersburg Polytechnic University, Russian Federation

M.Sc. in Electrophysics, St. Petersburg Polytechnic University, Russian Federation

Skills

- Programming Languages: Python, C/C++, MATLAB, Perl, Tcl
- Development environment: GitHub
- Languages: English, Hebrew, Russian

Publications and patents

1. Yan J., Aydiner A., **Messerman D.**, Zhu J., Camacho Mora A., Liao J., Ma M., Jiao D. Efficient Equalization Optimization Algorithm for Signaling Analysis. Nonlinear System 2019 IEEE MTT-S International Conference on Numerical Electromagnetic and Multiphysics Modeling and Optimization (NEMO).
2. Reisfeld G., **Messerman D.**, Bone N., Lazar A. VARAN: variability analysis for memory cell robustness. Proc. SPIE 6925, 69250L, 2008.
3. **Messerman D.**, Seltser M. Reorganizing rectangular layout structures for improved extraction. - United States Patent 7389001 B2, 2008.
4. Chakravarty S., Ben-Noon Y., Chiprout E., Mazumder M., **Messerman D.** Inductance Modeling. - United States Patent 7325208 B2, 2008.
5. **Messerman D.** Mathematical modeling of wave processes in high-voltage overhead power lines across a wide range of frequencies and soil electrical parameters. - Ph. D. Thesis, Moscow, 1989.

Recommendations

Ronen Cohen, Senior Staff Software Engineer

I had the privilege of managing Dmitry for several years, during which he provided exceptional support for various Analog tools and flows for teams in Israel. Dmitry is an outstanding CAD Engineer whom I strongly recommend. His unique blend of deep technical expertise and professionalism is evident in every tool and workflow he develops or maintains. He has a remarkable ability to translate complex requirements into practical, straightforward solutions, while clients greatly value his clear communication and can-do attitude. No matter how challenging the task, Dmitry was always the person I could rely on to deliver results.

Gideon Reisfeld, Ph.D. in Physics

Dmitry was my manager from 2006 to 2011. The technical depth and contributions I experienced under his leadership were truly exceptional. His expertise and innovative thinking in electrical engineering, mathematics, and computer science brought a transformative leap in the quality of the CAD solutions we developed.

Among his many achievements were evaluations in Static Timing Analysis within Intel's timing tool, the development of a thermal back-annotation tool for debugging tester thermal performance, and robust memory analysis techniques.

In short, if I were to choose someone as the ultimate debugger and a true technology maverick, Dmitry would be my first choice.